ELECTRONICGIANT

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EG1615 Chip Datasheet

Resonant Dual Active Bridge (DAB) Control Chip

Version Change Log

Version number	date	describe
V1.0	EG1615 Datasheet Draft, January 10, 2022.	
V2.0	On March 4, 2025, some descriptions of communication functions were modified.	

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EG1615 Chip Datasheet **V2.0**

1. Features

ÿ Employs an LC resonant dual active bridge (DAB) topology, enabling soft switching and high efficiency. ÿ Dedicated to the DC-to-DC section of bidirectional inverters, enabling inverter boost and charging management control. ÿ Integrates two 600V half-bridge high-voltage MOSFET drivers with a drive capability of ±2A; each driver has two independent MOSFET peaks.

The overcurrent protection circuit and the comparator with a built-in 200mV reference source allow users to set the overcurrent protection value.

ÿ Functions as an inverter/boost converter:

ÿ Supports externally adjustable PWM frequency, ranging from 40kHz to 150kHz, for adjustment during LC resonance. ÿ Supports PWM soft-start function with a soft-start time of 500ms. ÿ 50% duty cycle output, supporting adjustable dead time of 500ns, 700ns, 1µs, and 1.5µs. ÿ Output voltage feedback uses under-closed-loop control. ÿ High-voltage bus overvoltage protection. ÿ Battery overcurrent protection and transformer high-voltage overcurrent protection.

ÿ Battery overvoltage protection. ÿ Battery undervoltage protection. ÿ Over-temperature protection. ÿ Intelligent fan control. ÿ Functions as a battery charger:

• Fixed 50% duty cycle output, facilitating resonant soft-switching control. • Automatic synchronous rectification function, facilitating high-current charging and improving charger efficiency. • Combined with the PFC voltage regulation function in the EG8026, it can achieve constant current (CC) and constant voltage (CV) characteristics in the charging section. • Battery-side overcurrent protection and transformer high-voltage-side overcurrent protection. • Supports automatic charging stop function when fully charged. • Adjustable dead time: 500ns, 700ns, 1µs, 1.5µs. • Over-temperature protection. • Intelligent fan control.

Supports UART serial communication with a baud rate of 9600.

Depending on the customer's application, Yijing Microelectronics can modify the corresponding functions or parameters.

Operating power supply: +3.3V and +12V.

Package: LQFP64.

2. Description

The EG1615 chip is a control chip specifically designed for DC-to-DC inverter boost and battery charging management in bidirectional inverters (the same circuit can function as both an inverter and a battery charger). It integrates two 600V half-bridge high-voltage MOS drivers with an output current capability of +/-2A. It features four independent cycle-by-cycle PWM shutdown protections to effectively prevent damage to the MOS in extreme cases due to excessive peak current. Additionally, it provides two SD pins, SD1 and SD2. SD1 is the cycle-by-cycle shutdown pin for driver 1 HO1 and LO1, and SD2 is the cycle-by-cycle shutdown pin for driver 2 HO2 and LO2. Combined with an external comparator and SD functions, it can achieve overcurrent or short-circuit protection.

The EG1615's main functions consist of two parts. The first part is inverter boost control, which mainly converts the low-voltage DC voltage of the battery into a high-voltage DC voltage to supply the high voltage required by the subsequent DC/AC inverter. The second part is charger buck control, which mainly converts the high-voltage DC voltage of the mains power after being boosted by PFC into a low-voltage DC voltage to supply the constant voltage and constant current charging required for battery charging.

The EG1615 employs an LC resonant dual active bridge (DAB) topology, enabling bidirectional energy control. This, combined with the PWM operating frequency and LC resonant...

Matching the vibration parameters enables soft-switching control of the switching transistor, making it easy to apply to applications requiring high switching frequency and high efficiency.

When the EG1615 is used as an inverter boost converter, the duty cycle of the low-voltage side H-bridge is 50%, and the high-voltage side H-bridge uses the body diode inside the MOS for rectification output. The voltage feedback adopts a shallow closed-loop regulation mode (i.e., no-load maximum voltage limit, open-loop under load), which effectively prevents the voltage from being too high under no-load conditions, thus preventing device burnout. The resonant parameters are adjusted by an external LC. The chip supports PWM operating frequencies of 40K-150KHz. The current transformer connected in series on the high-voltage side can be used for output short-circuit protection.

When the EG1615 is used for charging and step-down, the duty cycle of the high-voltage side H-bridge is 50%. The low-voltage side H-bridge uses the internal diodes of the MOS for rectification output and the MOS tube for automatic tracking and synchronous rectification. The constant voltage (CV) and constant current (CC) of the charging need to be coordinated with the PFC voltage regulation function in the EG8026. The current transformer connected in series on the high-voltage side can be used for output short-circuit protection.

3. Application Areas

ÿ Bidirectional sine wave inverter ÿ Mobile energy
storage power supply ÿ Photovoltaic
inverter ÿ Uninterruptible power
supply (UPS) ÿ Battery energy wall

4. Pins

4.1 Pin Definitions

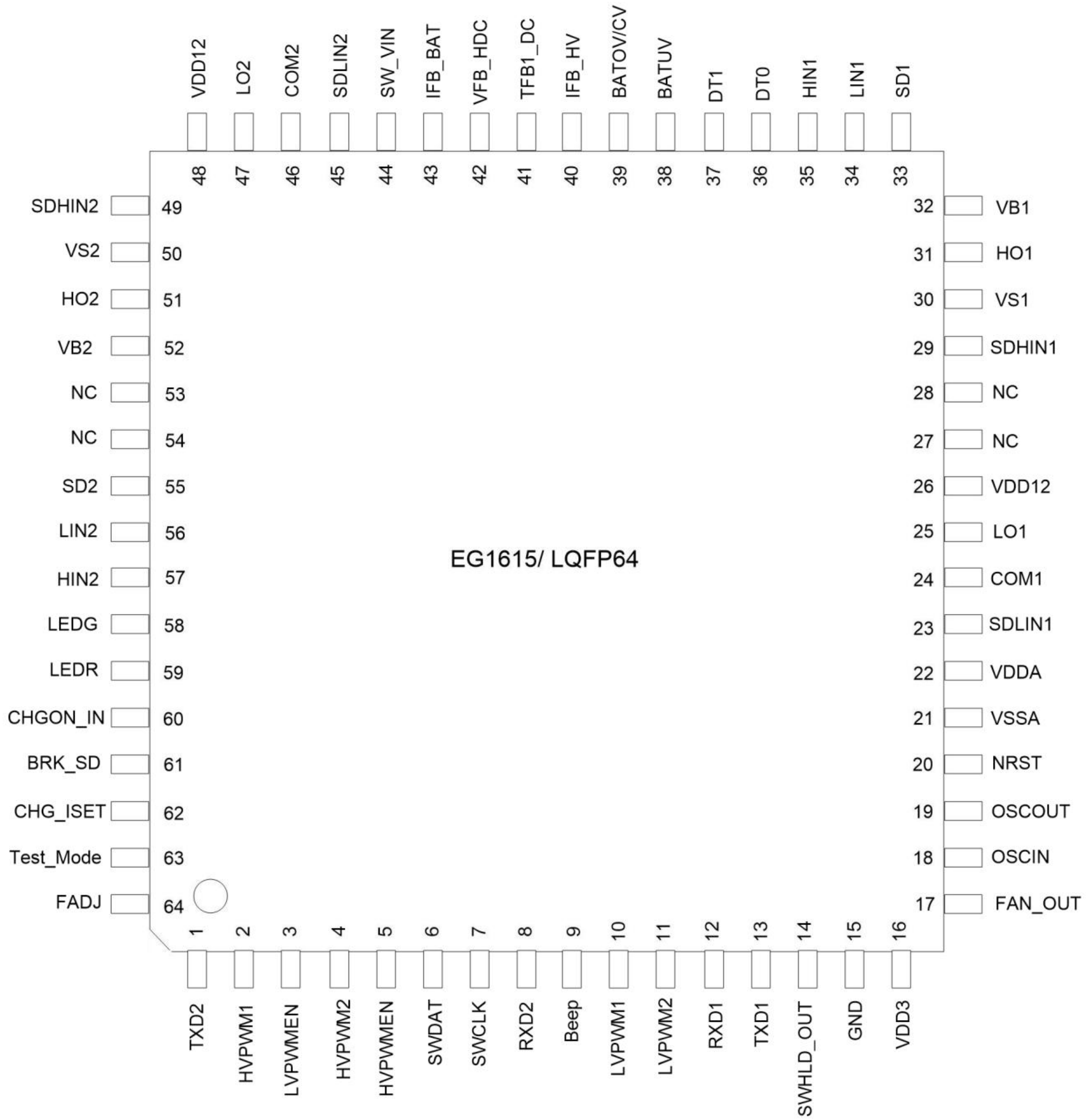

Figure 4-1. EG1615 Pin Definitions

4.2 Pin Description

Pin order Number	Pin Name **I/O**		describe
1	TXD2		Serial Port 2 Communication Data Transmitter
2	HVPWM1		O is used for push-pull output PWM1 of the H-bridge on the high-voltage side of the transformer.
3	LVPWMEN		The PWM enable control pin of the low-voltage side H-bridge , "1" enables LVPWMx output, "0" disables it. Disable LVPWMx output
4	HVPWM2		O is used for push-pull output PWM2 of the H-bridge on the high-voltage side of the transformer.
5	HVPWMEN		The PWM enable control pin of the high-voltage side H-bridge : "1" enables HVPWMx output, "0" disables HVPWMx output.
6	SWDAT		Firmware upgrade data
7	SWCLK		O Firmware upgrade clock
8	RXD2		Serial Port 2 is the data receiving pin for communication (this pin cannot be left floating; it needs to be connected to a pull-up resistor, such as 10K to 3.3V).
9	Beep		O Buzzer alarm output, active high level
10	LVPWM1		O is used for push-pull output PWM1 of the H-bridge on the low-voltage side of the transformer.
11	LVPWM2		O is used for push-pull output PWM2 of the H-bridge on the low-voltage side of the transformer.
12	RXD1		Serial Port 1 is the data receiving pin for communication (this pin must not be left floating; a pull-up resistor, such as 10KV to 3.3V, is required).
13	TXD1		Serial port 1 communication data sending end
14 SWHLD_OUT			O Power on/off latch signal output, for push-button switch applications.
15	GND		GND chip ground terminal
16	VDD3		The +3.3V power supply terminal of the Power chip
17	FAN_OUT		O Fan control output, active high.
18	OSCIN		1 8M crystal oscillator pin 1
19	OSCOUT		O 8M crystal oscillator pin 2
20	NRST		I. Chip reset pin: Low level is effective for reset.
21	VSSA		The analog ground terminal of the GND chip
22	VDDA		The analog power supply terminal of the Power chip is +3.3V.
23	SDLIN1		The low-side MOSFET peak current protection input of driver 1 has an internal reference voltage of 200mV.
24	COM1		GND power ground of driver 1
25	LO1		Low-side gate drive output of driver 1
26	VDD12		The power supply for Power Driver 1 has an input voltage range of 10V-20V.
27	NC		- Empty pin, used for high-voltage isolation
28	NC		- Empty pin, used for high-voltage isolation
29	SDHIN1		The high-side MOSFET peak current protection input of driver 2 has an internal reference voltage of 200mV.
30	VS1		High-end floating output of driver 1
31	HO1		High-side gate drive output of driver 1
32	VB1		The floating power supply for Power Driver 1 requires an external 10uF bootstrap capacitor.
33	SD1		SD control terminal of drive 1
34	LIN1		The low-side control signal input terminal of driver 1 controls the on/off state of the low-side power MOSFET.
35	HIN1		The high-side control signal input terminal of driver 1 controls the on/off state of the high-side power MOSFET.

36	DT0	I. Settings via serial port
37	DT1	
38	BATUV	I. Battery voltage undervoltage detection, undervoltage alarm, BATUV < 1.65V; undervoltage shutdown, BATUV < 1.6V.
39	PISTON/CV	Battery voltage overvoltage detection and constant charging voltage output; overvoltage alarm and shutdown for BATOV > 2.5V; constant charging voltage output. The voltage reference value is 2.4V.
40	IFB_HV	I. Transformer high-voltage side current feedback input pin: IFB > 0.1V, fan on; IFB > 0.6V, overcurrent off.
41	TFB1_DC	Temperature sensing pin 1: Output off when TFB < 1.15V, output resumes when TFB > 1.28V.
42	VFB_HDC	I. High-voltage side DC voltage feedback, via optocoupler isolation. When VFB > 3V, the output is shut off.
43	IFB_BAT	I. Battery current feedback input pin
44	SW_VIN	I. Power On/Off Button Voltage Detection Input Pin
45	SDLIN2	The low-side MOSFET peak current protection input of driver 2 has an internal reference voltage of 200mV.
46	COM2	GND driver 2 power ground
47	LO2	Low-side gate drive output of driver 2
48	VDD12	The power supply section of Power Driver 2 has an input voltage range of 10V-20V.
49	SDHIN2	The high-side MOSFET peak current protection input of driver 2 has an internal reference voltage of 200mV.
50	VS2	High-end floating output of driver 2
51	HO2	High-side gate drive output of driver 2
52	VB2	The floating power supply for Power Driver 2 requires an external 10uF bootstrap capacitor.
53	NC	- Empty pin, used for high-voltage isolation
54	NC	- Empty pin, used for high-voltage isolation
55	SD2	SD control terminal of drive 1
56	LIN2	The low-side control signal input terminal of driver 2 controls the on/off state of the low-side power MOSFET.
57	HIN2	The high-side control signal input terminal of driver 2 controls the on/off state of the high-side power MOSFET.
58	LEAD	O Operation LED indicator output
59	LEATHER	O Fault LED indicator output
60	CHGON_IN	I. Charging buck mode enable signal input
61	BRK_SD	I Short circuit protection
62	CHG_ISET	I. Settings via serial port
63	Test_Mode	I. Settings via serial port
64	FADJ	I. Settings via serial port

5. Structural Diagram

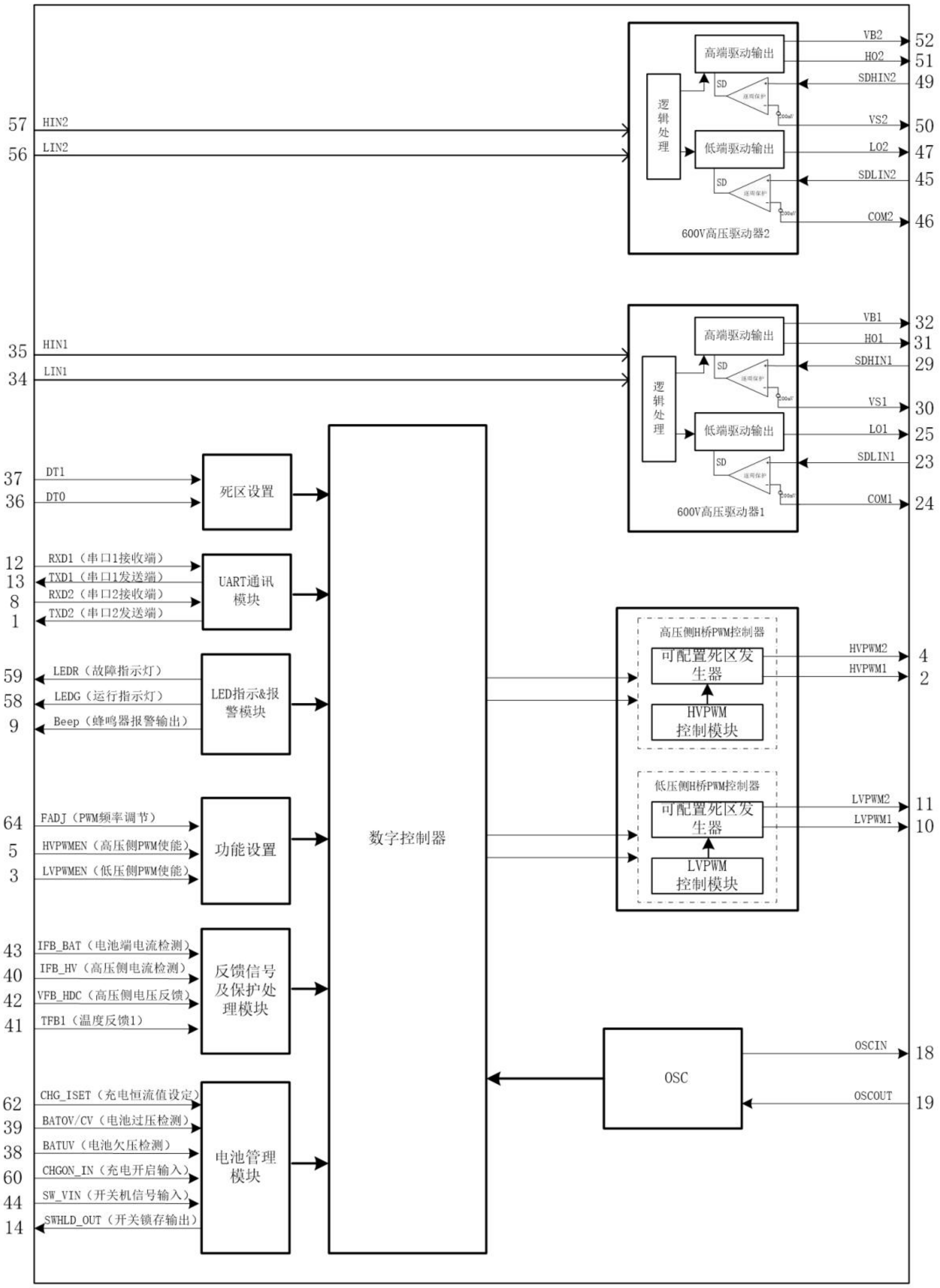

Figure 5-1. EG1615 Structural Diagram

6. Typical Application Circuits

6.1 Schematic diagram of EG1615 DC/DC control board application

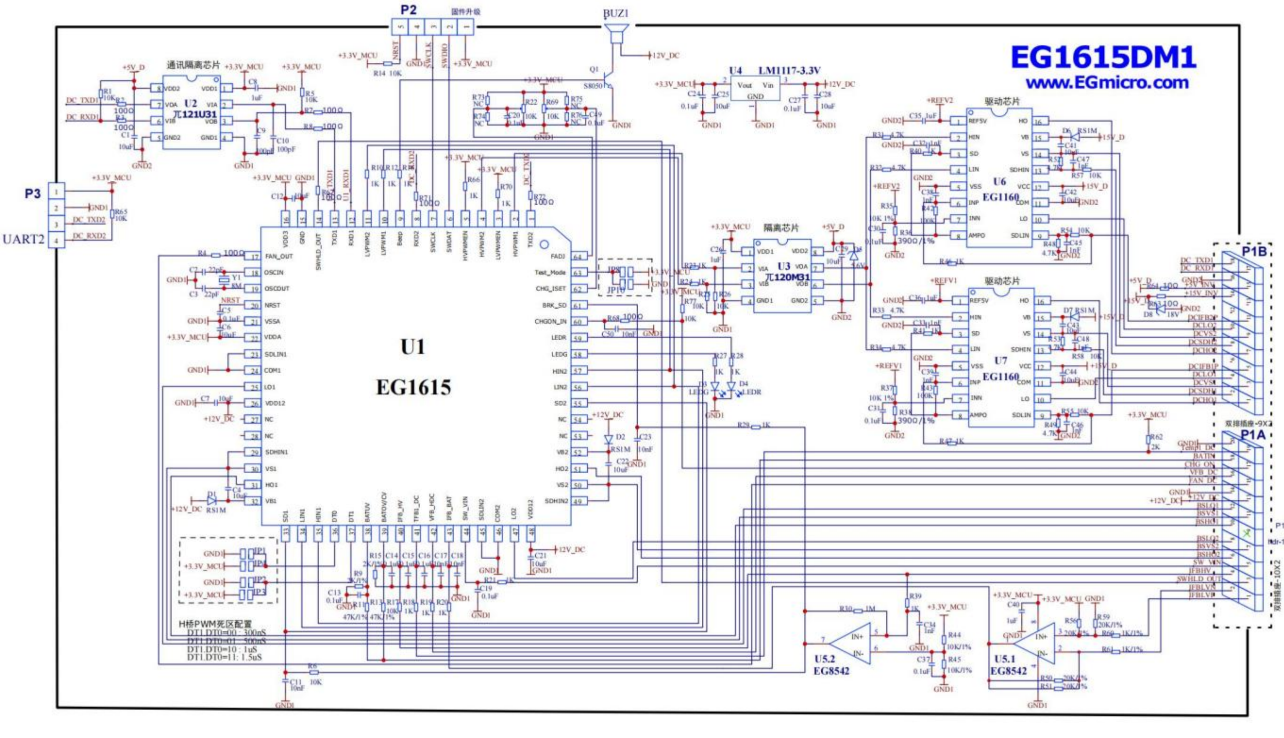

Figure 6-1. Schematic diagram of EG1615 control board

6.2 EG1615 Motherboard Application Schematic

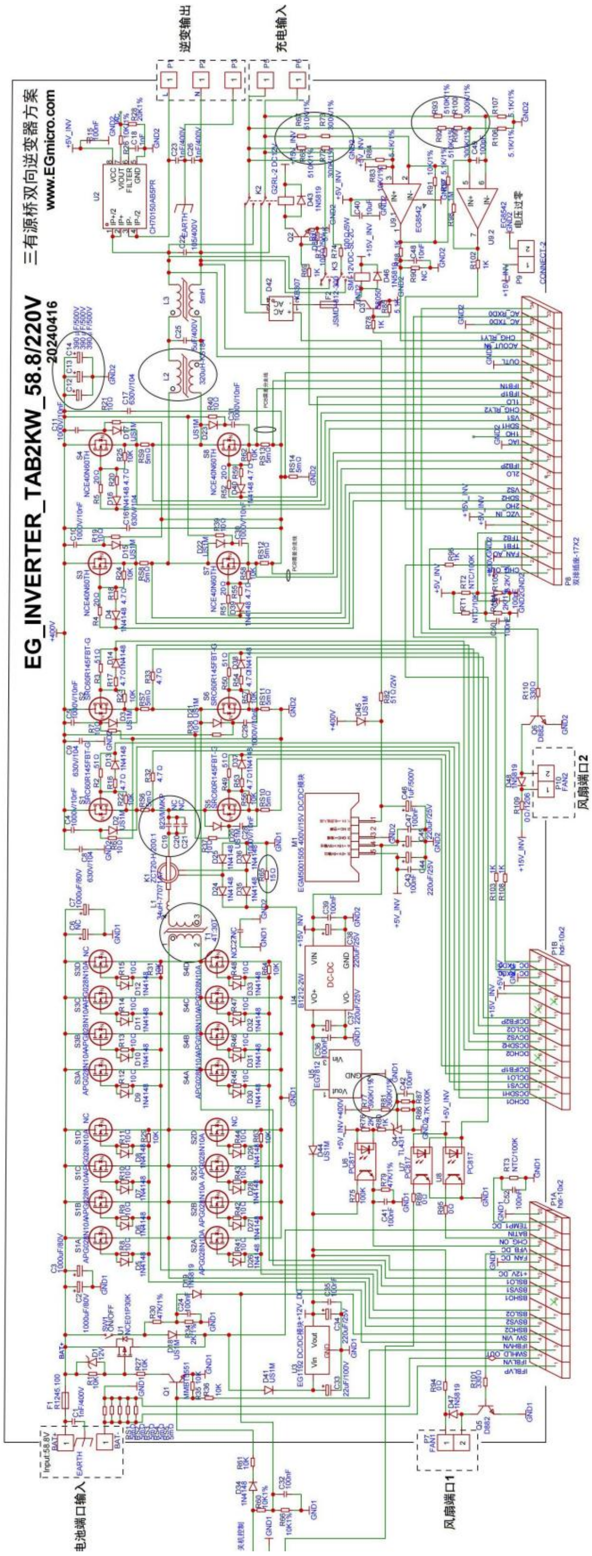

Figure 6-2. Schematic diagram of EG1615 motherboard application

6.3 Schematic diagram of EG8026 DC/AC control board

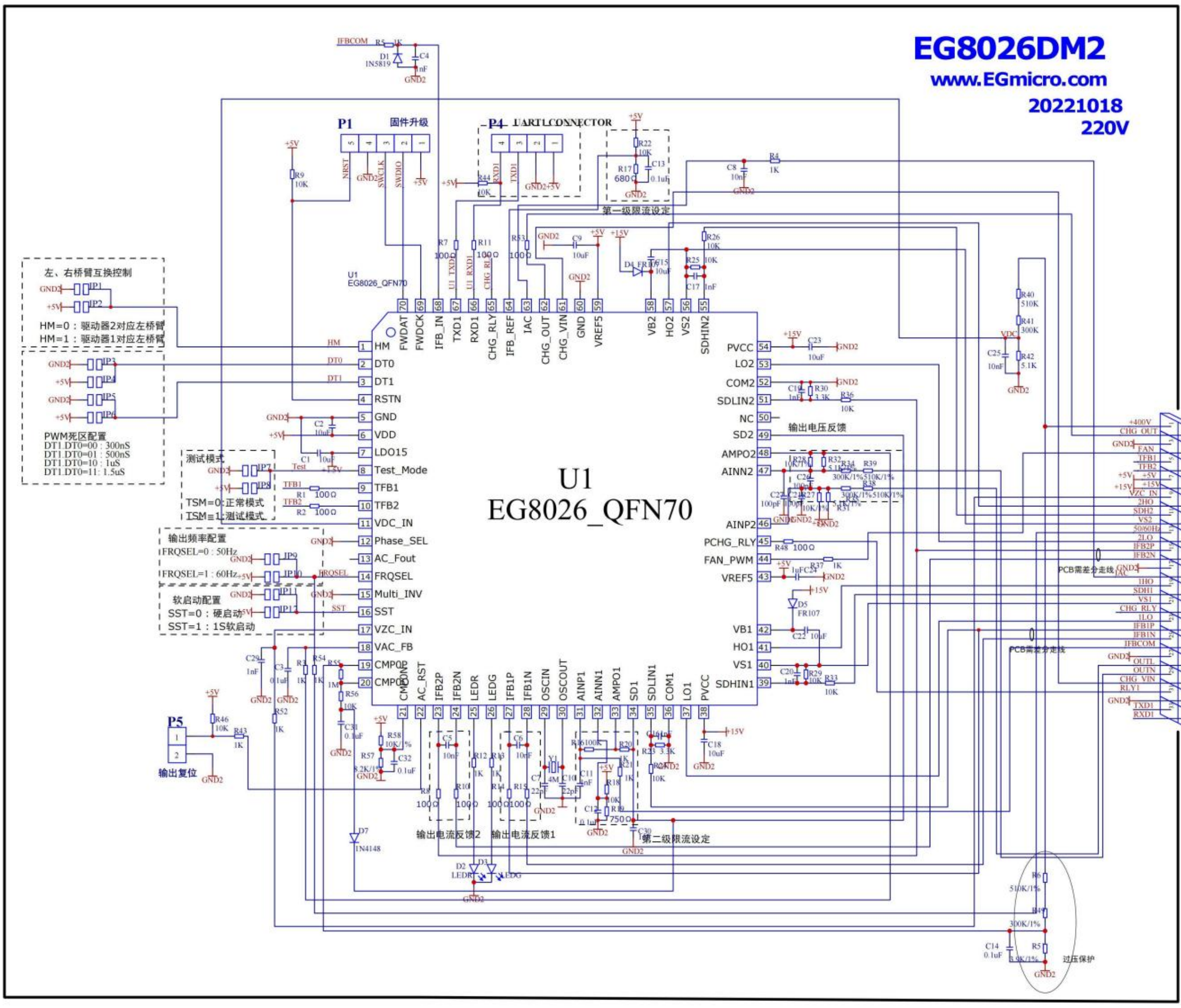

Figure 6-4. Schematic diagram of EG8026 control board

7. Electrical characteristics

7.1 Limiting Parameters

Unless otherwise specified, under the condition of TA=25ÿ

Symbolic parameter	name	Test conditions	Minimum and	maximum	units
VDD3ÿVDDA	power supply	The VDD3 and VDDA pins relative to GND Voltage	-0.3	4.0	In
VDD12	Driver power supply	The voltage of the VDD_12V pin relative to GND. Pressure	-0.3	20	In
VB1ÿVB2	Bootable high-end VB power supply		-0.3	600	In
VS1ÿVS2	High-end power supply		VB -20	VB+0.3	In
HO1ÿHO2	High-end drive output		VS -0.3	VB+0.3	In
LO1ÿLO2	Low-end drive output		-0.3	VDD12	In
SDHIN1, SDHIN2	High-side SD comparator positive input terminals		VS-0.3	VS+5	In
SDLIN1, SDLIN2	are the low-end SD comparator inputs (positive terminals).		-0.3	5.5	In
SD1, SD2	SD logic control input		-0.3	5.5	In
HIN1ÿHIN2	High-channel logic signal input level		-0.3	VDD12	In
LIN1, LIN2	Low channel logic signal input level		-0.3	VDD12	In
I/O	General purpose input/output ports	The voltage of a general-purpose I/O pin to GND	is -0.3.	4.0	In
Isink	Maximum output sink current of the output pin		-	25	m.a.
Isource	Maximum output pull-up current of the output pin		-	-25	m.a.
FACING	Ambient temperature		-40	105	ÿ
Tstr	Storage temperature		-65	125	ÿ

Note: Exceeding the listed limits may cause permanent damage to the chip's internal components. Prolonged operation under extreme conditions may affect the chip's reliability.

7.2 Typical Parameters

Unless otherwise specified, at TA=25ÿ, OSC=8MHz

symbol	Parameter Name	Test Conditions	Minimum	Typical	Maximum	Unit
VDD3ÿVDDA	power supply	-	2.4	3.3	3.6	In
VDD12	Driver power supply	-	10	12	18	In
Ivdd12	VDD12 static electricity flow	VDD12=12V	-	1	1.5	m.a.
Ivdd3 + Ivdda	VDD quiescent current	VDD=3.3V	-	5	20	m.a.
Internal high-voltage **MOS** driver						
VB1-VS1ÿVB2-VS2	High-end drive supply Electric power supply	VB voltage relative to VS terminal	10	12	18	In
SDHIN1ÿSDHIN2	High-end current comparison Internal reference of the instrument	Relative VS voltage	-	200	-	mV
SDLIN1, SDLIN2	Low-end current comparison Internal reference of the instrument	Relative COM terminal voltage	-	200	-	mV
SD1, SD2	SD logic control input enter	voltage relative to GND	0	-	5	In
IO+	IO output current pulling ability	-	1.8	2	-	A
IO-	IO output sink current ability	-	2	2.5	-	A
feedback						
VFB_HDC	Peak Feedback Benchmark Voltage	-	-	>3.0	-	In
BATUV	Battery undervoltage protection	-	-	<1.6	-	In
PISTON/CV	Battery overvoltage protection	-	-	>2.5	-	In
IFB_HV	Current protection reference Voltage	-	-	>0.6	-	In
	Overcurrent detection delay time	-	-	10	-	mS
	Fan start voltage	-	-	>0.1	-	In
SW_VIN	Power switch voltage Detection	-	-	>1	-	In
TFB1_DC	Over-temperature protection value	-	-	<1.15	-	In
	Exit overtemperature protection value	-	-	>1.28	-	In
	Fan on value	-	-	<2.3	-	In

	Fan off value	·	·	>2.4	·	In
Function settings						
FADJ	FADJ pin input power Pressure range	·	0	·	3.3	In
	Corresponding PWM frequency	·	40	·	150	KHz
CHG_ISET	CHG_ISET Foot input Input voltage range	·	0	·	3.3	In
	The corresponding charging current is Rsense = 1mÿ, A = 10		0	·	165	A
UART communication port						
RXD1ÿTXD1 RXD2ÿTXD2	Wine(H) Input high potential	VDD3=3.3V, IOH=-10mA	2.3	3.3	·	In
	Wine(L) Input low potential	VDD3=3.3V, IOL=10mA	·	0	0.5	In
Control module and indicator module						
LEDR, LEDG, Beep, FAN_OUT, SWHLD_OUT	Vout(H) Output high potential	VDD=3.3V, IOH=-10mA	2.3	3.3	·	In
	Vout(L) Output low potential	VDD=3.3V, IOL=10mA	·	0	0.3	In
FADJ, HVPWMEN, LVPWMEN, DT0ÿDT1ÿ Test_Mode	Wine(H) Input high potential	VDD=3.3V	2.3	3.3	·	In
	Wine(L) Input low potential	VDD=3.3V	-0.3	0	0.5	In

8. Application Design

8.1 Main Topology of Bidirectional Inverter

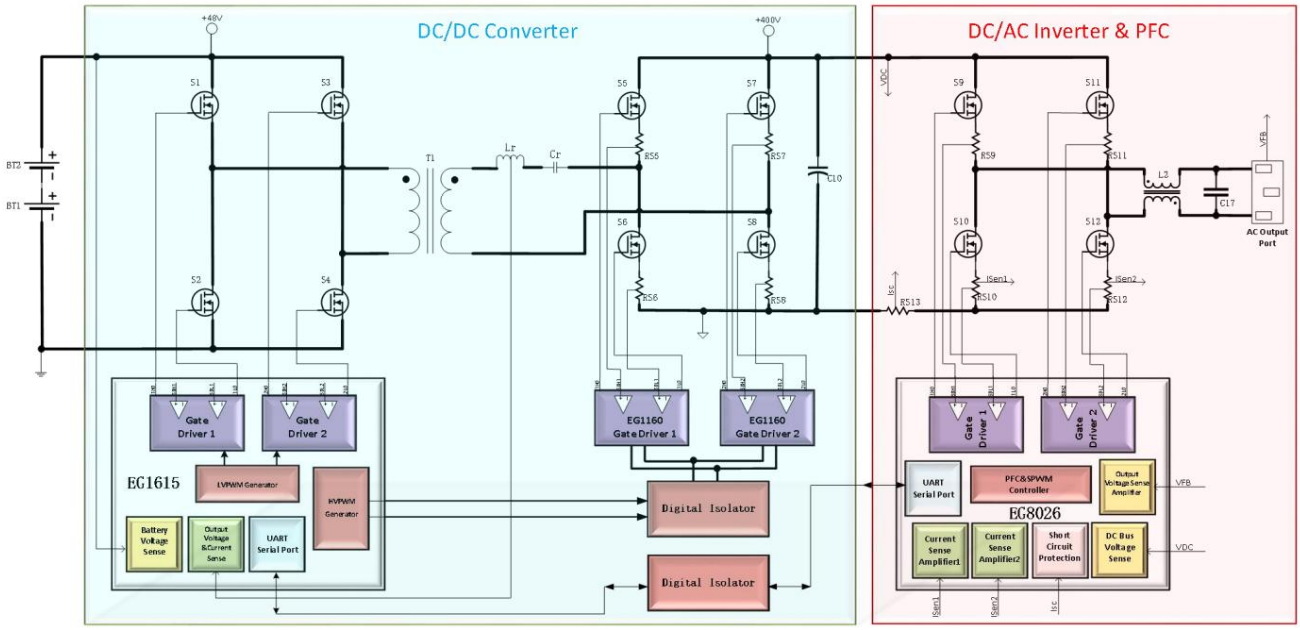

Figure 8.1 Main topology diagram of EG1615 bidirectional inverter

The main topology of the bidirectional inverter solution developed by Yijing Microelectronics adopts a Triple Active Bridge (TAB) circuit structure, as shown in Figure 8.1. The EG1615 chip in the figure is

responsible for controlling the DC/DC conversion section, employing an LC resonant dual active bridge (DAB) topology. The circuit, shown in the DC/DC Converter section of Figure 8.1, enables bidirectional energy transfer. When acting as an inverter boost converter, it converts the low-voltage DC voltage from the battery into a high-voltage DC voltage to supply the high voltage required for the subsequent DC/AC inverter. When acting as a charger buck converter, it converts the high-voltage DC voltage from the mains power boosted by PFC into a low-voltage DC voltage to supply the constant voltage and constant current charging required for battery charging. Combined with the matching of LC resonant parameters and PWM operating frequency, the DC/DC stage can achieve soft-switching control of the switching transistors.

The EG8026 chip in the diagram is responsible for controlling the DC/AC inverter and PFC boost section. When performing DC/AC inverter operation, the EG8026 uses a center-aligned PWM modulation method with a modulation frequency of 20kHz. The advantage of this modulation method is that the switching frequency of the H-bridge transistors is 20kHz, and the switching frequency on the output inductor and output capacitor is twice the PWM frequency (40kHz). Compared with the unipolar or bipolar modulation methods of traditional inverters, at the same power, the switching losses on the MOSFETs or IGBTs are the same, while the frequency acting on the output inductor and capacitor is twice that of the traditional method. This modulation method can reduce the size and wire diameter of the inductor. When performing PFC boost operation, the EG8026 uses a traditional bridgeless PFC circuit structure with two lower-side switching transistors of the active bridge. S10 and S12 are used for PWM modulation. The two power supply switches S9 and S11 use internal body diodes for synchronous freewheeling. The modulation frequency is 20KHz, and the average current control algorithm is used for control.

8.2 LC Resonance Parameters

As shown in the circuit structure in Figure 8-1, the resonant circuit consists of Lr and Cr in series resonance. The resonant frequency is calculated using the formula f=1/[2ÿÿ(LrCr)], where Lr includes the leakage inductance of the transformer. The PWM frequency is adjusted to equal the LC resonant frequency, making it correspond to the resonant point of Lr and Cr. This aims to enable the S1, S2, S3, and S4 switches to operate in ZCS soft-switching mode, achieving high frequency and high efficiency in applications. Lr should be selected using a low-permeability magnetic ring with strong anti-saturation capability, such as an iron-silicon-aluminum magnetic ring. Cr should be selected using a capacitor capable of withstanding high current, high voltage, and low temperature rise, such as an MMKP capacitor.

8.3 Self-locking circuit for power on/off buttons

To achieve zero-power voltage sampling at the battery terminal and power-on/off functionality without a self-locking button, the EG1615 uses three main pins to implement this function. Yes, the pins are 44 (SW_VIN), 60 (CHGON_IN), and 14 (SWHLD_OUT), respectively. The circuit structure is shown in Figure 8.2.

Pin 44 (SW_VIN) is used for button operation detection. When the voltage on this pin is greater than 1V, it is determined that a button operation signal has been input. The foot has a 10ms anti-shake filter.

Pin 14 (SWHLD_OUT) is used as the switch latching signal output. When the SW_VIN signal is determined to be valid, SWHLD_OUT will output... A high level is used for the self-locking switch circuit. When the button is pressed again and the SW_VIN signal is valid, SWHLD_OUT will output a low level to disable the self-locking circuit.

Pin 60 (CHGON_IN) is used to activate the SWHLD_OUT output and the battery sampling signal during charging.

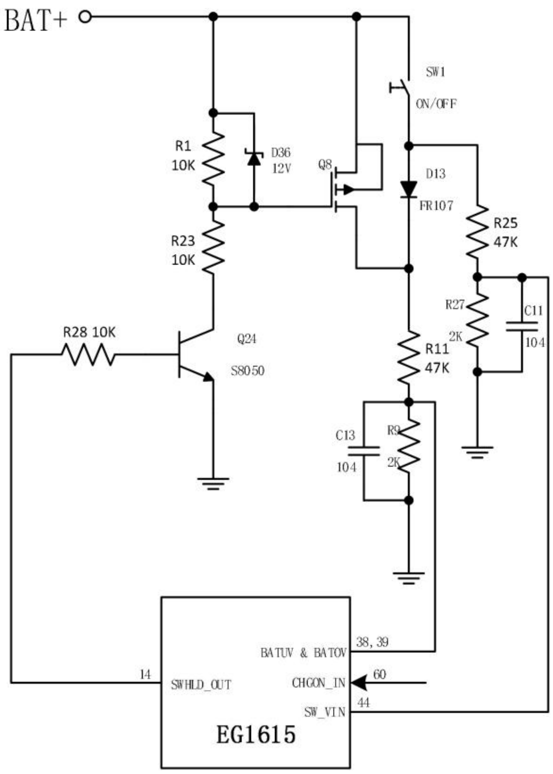

Figure 8.2 Circuit for power on/off without self-locking button

8.4 Battery undervoltage shutdown and undervoltage buzzer during inverter operation

When the bidirectional inverter is operating in inverter mode, to prevent damage caused by over-discharge of the battery, the EG1615 chip has a built-in battery voltage detection circuit, providing two protection functions: battery undervoltage alarm and battery undervoltage shutdown, as shown in Figure 8.3. The battery voltage is detected by using an external voltage divider resistor connected to pin 38 of the EG1615. The chip's internal undervoltage shutdown voltage comparison value is 1.60V, the undervoltage alarm voltage comparison value is 1.65V, and the undervoltage release voltage comparison value is 1.75V.

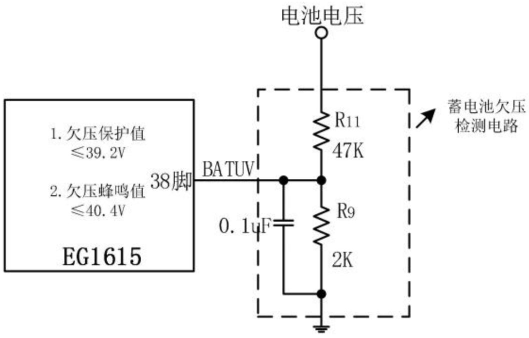

Figure 8.3 EG1615 Battery Undervoltage Detection Circuit

Figure 8.3 shows the recommended external voltage divider resistor values for a 13-cell lithium battery application. R11 (47K) and R9 (2K) together form a 40.4V undervoltage alarm. The system features an undervoltage shutdown protection function with a voltage of 39.2V. To modify the undervoltage protection value to another value, refer to the formula UBATUV=(1 + R11/R9) x 1.60V, where UBATUV is the battery undervoltage protection value. For example, in a 7-cell lithium battery application, if a battery undervoltage shutdown value of 21V is desired, using the above formula, R9 can be first selected as 2K, and then R11 can be calculated as 24.25K.

8.5 Battery overvoltage shutdown during inversion and battery voltage sampling during charging

When the bidirectional inverter is operating in inverter mode, to prevent damage to the inverter due to excessively high voltage when a mismatched battery is connected, the EG1615 chip has a built-in battery overvoltage shutdown and buzzer alarm protection function, as shown in Figure 8.4. The battery overvoltage detection is achieved by using an external voltage divider resistor connected to pin 39 of the EG1615. The internal overvoltage shutdown and alarm voltage comparison value of the chip is 2.5V.

Figure 8.4 shows the recommended external voltage divider resistor values for a 13-cell lithium battery application. R13 (47K) and R15 (2K) form a 61.25V overvoltage shutdown and overvoltage buzzer alarm. If you need to modify it to other overvoltage protection values, you can refer to the formula UBATOV=(1 + R13/R15) x 2.5V, where UBATOV is the battery overvoltage protection value. For example, in a 7-cell lithium battery application, if you want the battery overvoltage shutdown value to be 30V, according to the above formula, you can first select R15 as 2K, and then calculate R13 as 22K.

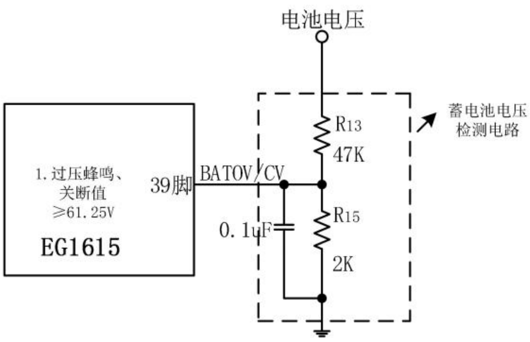

Figure 8.4 EG1615 Battery Overvoltage Detection Circuit

When the bidirectional inverter is operating in charging mode, pin 39 of the EG1615 is used for battery voltage sampling and constant voltage feedback. The internal reference voltage is 2.4V. When setting it, refer to the formula UBATCV = (1 + R13/R15) x 2.4V, where UBATCV is the constant voltage charging voltage. As shown in Figure 8.4, the constant voltage value is UBATCV = 58.8V. To fine-tune the CV constant voltage value, add a resistor to R13.

connected in series for fine-tuning. Adjusting the voltage divider resistor ratio will cause the battery overvoltage protection value to deviate. Generally, overvoltage protection is to prevent mismatched batteries from being connected to the inverter, and the fine-tuned voltage value usually does not affect the overall performance of the inverter. If you do not want to adjust the CV constant voltage value by modifying hardware parameters, Yijing Microelectronics provides host computer software for users to modify the CV constant voltage value. Users can download the corresponding host computer software from our website or by contacting us.

8.6 IFB_BAT Battery Terminal Current Detection and Constant Current Feedback During Charging

The EG1615 uses a high-precision alloy resistor to sample the battery terminal current. The circuit structure is shown in Figure 8.5. The current on the RS1 resistor is amplified by an external operational amplifier by 10 times and then sent to pin 43 of the EG1615 for current detection and various function processing.

The main functions implemented by the IFB_BAT pin include detecting charging current and discharging current, setting the charging constant current value CC, and setting the conditions for enabling synchronous freewheeling of the low-voltage H-bridge during charging.

During charging and discharging current detection, the EG1615 uses the sampled current for power calculations, protection, fan on/off, and synchronization.

Enabling and disabling freewheeling.

The following are operational amplifier configuration calculation formulas for reference:

Step 1: Calculate DC offset: op-amp output Vout_DC = $\frac{59}{56+59}$ x 3.3V = 20K / 40K x 3.3V ÿ 1.65V (Refer to Figure 8.2a)

Step 2: Calculate the op-amp amplification factor: A = (R50//R51)/(R61) = 10 Step 3: Calculate the op-amp output voltage: Vout = Vout_DC + A*Vin (Vin is the voltage across the sampling resistor RS1)

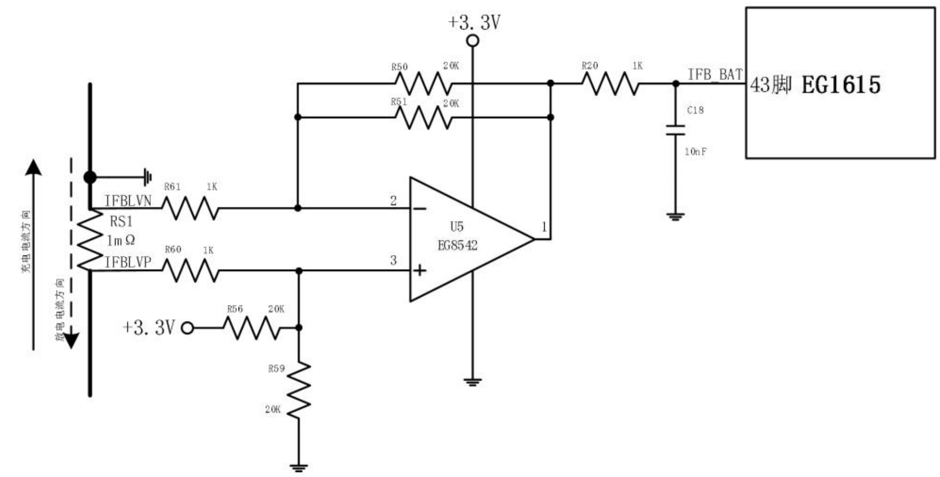

Figure 8.5 EG1615 Battery Terminal Current Detection Circuit

When routing PCB traces, IFBLVP and IFBLVN need to be run as differential signal lines. The maximum current is calculated as: Imax = (3300mV - 1650mV) / 10 / Rs. When Rs is set to 0.001ÿ, the maximum current Imax = 1650mV/10/0.001ÿ = 165A.

When applying this feature, the IFB_BAT pin must not be left floating or grounded. It must be connected strictly according to the method described in section 8.5. Otherwise, it will affect the constant current charging function, synchronous freewheeling, power protection and other functions, and may even lead to the risk of burning out components in severe cases.

8.7 IFB_HV High-voltage side overcurrent protection

The EG1615 uses transformer secondary current detection to achieve overcurrent protection shutdown function. The circuit structure uses a current transformer, as shown in Figure 8.7. The output voltage of T2 secondary is rectified by D23~D26 and supplied to the load resistor R32. This voltage is provided to pin 40 IFB_HV of the EG1615 for overcurrent judgment. When the voltage at pin IFB is greater than 0.6V, the overcurrent protection is activated, and the MOSFET output is turned off.

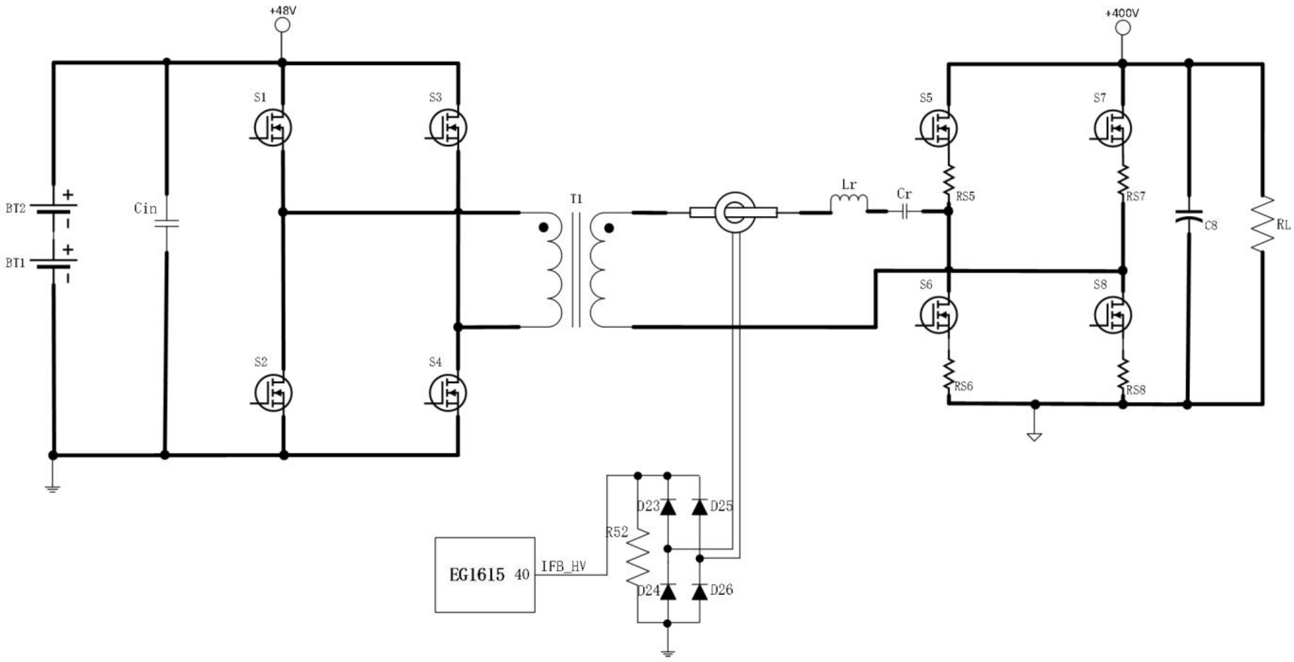

Figure 8.7 EG1615 Overcurrent Detection Circuit

8.8 Low-voltage side synchronous freewheeling

When performing buck charging, the low-voltage side H-bridge will be responsible for the low-voltage side rectification output. In order to support high-current fast charging, the EG1615 integrates a low-voltage side automatic synchronous freewheeling function. The activation and deactivation of the synchronous freewheeling is determined by the voltage of the IFB_BAT pin. When working in charging mode, as long as the IFB_BAT pin is detected to be greater than 50mV, the EG1615 will activate the low-voltage side synchronous freewheeling to improve the efficiency of the converter. At the same time, a 20mV hysteresis voltage is applied to prevent oscillation near the activation point.

8.9 TFB1_DC Temperature Detection Feedback

The EG1615 pin TFB1_DC measures the inverter's operating temperature, primarily for over-temperature protection. The circuit structure is shown in Figure 8.8. An NTC thermistor RT3 and measuring resistor R62 form a simple voltage divider circuit. The voltage division value changes with temperature, reflecting the value of the NTC resistor and thus the corresponding temperature. The NTC thermistor is selected with a resistance of 10Kÿ (B constant value is 3950ÿ) corresponding to 25ÿ. The over-temperature voltage of the TFB1_DC pin is set at 1.15V. When TFB1_DC < 1.15V , over-temperature protection occurs, and LVPWM1 and LVPWM2 output a low level to turn off the power MOSFET. When TFB_DC > 1.28V, the EG1615 exits over-temperature protection, and the inverter operates normally. When R62 is 2Kÿ, the corresponding over-temperature protection is at 85 degrees Celsius, and the exit point is 80 degrees Celsius. If over-temperature protection is not used, this pin needs to be pulled up to 3.3V.

Meanwhile, the TFB1_DC pin has a fan on/off function. When the voltage of the TFB1_DC pin is less than 2.3V and the corresponding temperature is greater than 45 degrees, the fan will turn on. When the voltage of the TFB1_DC pin is greater than 2.4V and the corresponding temperature is less than 40 degrees, the fan will turn off.

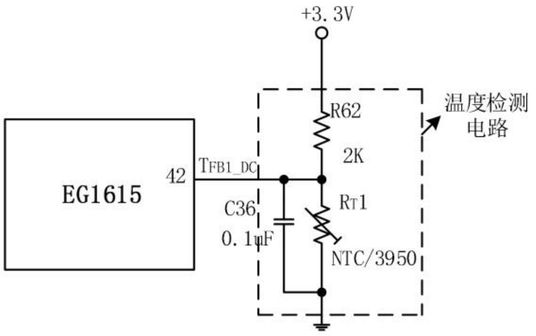

Figure 8.8 EG1615 Temperature Detection Circuit

8.10 VFB_HDC High-voltage side bus DC voltage feedback

The EG1615 pin VFB_HDC (pin 42) is used for DC voltage feedback on the high-voltage side bus during inverter boost. VFB_HDC uses under-closed pins. The loop control mode includes maximum voltage limiting under no-load conditions and open-loop output under load. When the voltage at the VFB_HDC pin exceeds 3V in no-load mode, the EG1615 will enter hiccup mode to limit the output voltage, effectively preventing overvoltage and potential device burnout under no-load conditions. When operating in load mode, the EG1615 will enter open-loop mode, and the VFB_HDC pin will be disabled without feedback. Similarly, during buck charging, the VFB_HDC pin remains disabled without any processing.

Figure 8.9 shows the DC voltage feedback circuit of the high-voltage side bus. The internal reference voltage of pin 1 Ref of TL431 is 2.5V. When setting the maximum output voltage, you can refer to the formula UHVMAX=[1+(R60+R64)/(R67//R68)] x 2.5V, where UHVMAX is the maximum output voltage value. As shown in the parameters of Figure 8.9, the maximum output voltage value can be obtained as UHVMAX=420V.

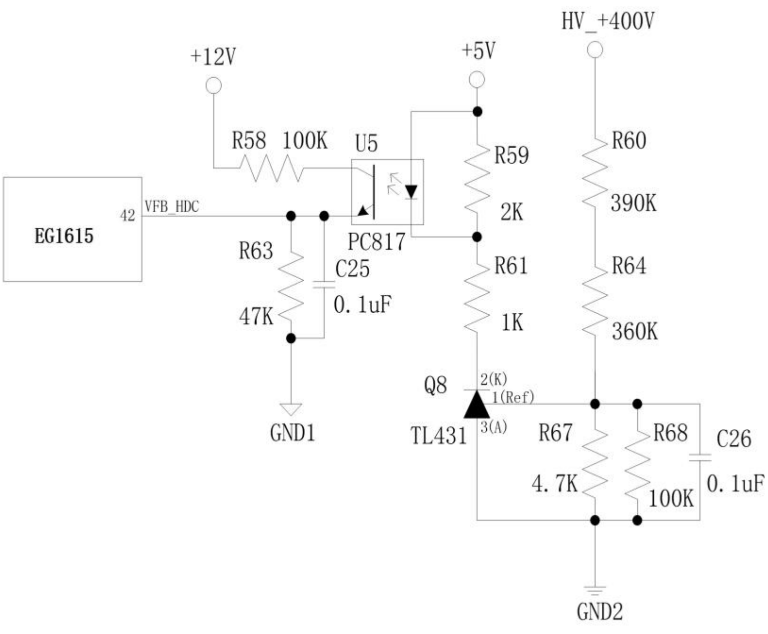

Figure 8.9 EG1615 VFB_HDC High-voltage side bus DC voltage feedback circuit

8.11 Dead Time

The dead time of the EG1615 can be set via serial port, with a default value of 500ns. Dead time control is one of the important parameters of power MOSFETs. If there is no dead time or the dead time is too small, the upper and lower power MOSFETs will conduct simultaneously, causing the MOSFETs to burn out. If the dead time is too large, it will cause waveform distortion and severe overheating of the power transistors.

8.12 FAN Fan Control

The EG1615 pin FAN_OUT (pin 17) is used to control the fan's on/off state, as shown in Figure 8.12. An external D882 transistor is connected to it.

A 330ÿ base resistor is used to drive a high-current fan. The

fan starts under the following conditions; any one condition must be met to turn on the fan, while all conditions must be met to turn it off: ÿ The voltage at

the IFB_BAT pin is greater than 50mV; ÿ The voltage at the IFB_HV pin is greater than 0.1V; ÿ

The voltage at the TFB1_DC pin is less than 2.3V ;

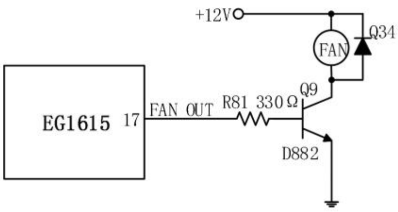

Figure 8.12 FAN fan control output

8.13 Beep Control

The EG1615's Beep pin (pin 9) is used to control the buzzer alarm, as shown in Figure 8.13. An external S8050 transistor and a 1K base resistor are connected to drive the buzzer. When the BATUV pin detects low battery voltage, the Beep pin outputs a high level, causing the buzzer to sound continuously. When the BATUV pin detects overvoltage, the Beep pin outputs a high level, producing two beeps lasting 300ms, repeating after 3 seconds. When the inverter is powered on or mains power is connected, the Beep pin outputs a high level, producing one beep lasting 200ms. When the inverter is powered off or charging is complete, the Beep pin outputs a high level, producing two beeps lasting 300ms.

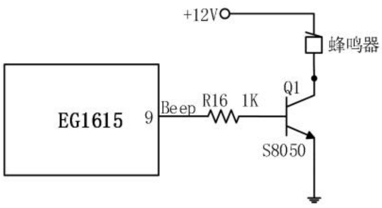

Figure 8.13 Buzzer control output

8.14 LED Operation and Fault Indication

LED Indicators During Inverter Operation: ÿ

Solid green LED, solid red LED: This indicates the inverter is operating normally and without faults. ÿ Flashing red LED, solid green

LED: This indicates the inverter has entered fault protection mode. The number of flashes can be used to determine the fault protection mode.

Determine the fault type. The specific definitions for the red LED are

as follows: ÿ Normal:

Always off ÿ Short circuit: Flashes once, then off for 2 seconds,

continuously ÿ Overcurrent: Flashes twice, then off for 2 seconds,

continuously ÿ Overvoltage: Flashes three times, then off for 2 seconds,

continuously ÿ Undervoltage: Flashes four times, then off for 2 seconds,

continuously ÿ Overtemperature: Flashes five times, then off for 2 seconds, continuously ÿ

LED indicator when the charger is in operation

• Red LED off, green LED flashing: This indicates charging is in progress. • Green LED constantly on, red LED constantly off: This indicates the battery is fully charged.

8.15 High-voltage side PWM enable and low-voltage side PWM enable control

The EG1615 supports independent PWM enable control on both the high-voltage and low-voltage sides. HVPWMEN is the PWM enable control pin for the high-voltage H-bridge, primarily used to enable or disable buck charging functions. "1" enables HVPWMx output, and "0" disables it. When the HVPWMEN enable pin is disabled, the charging function will be ineffective. LVPWMEN is the PWM enable control pin for the low-voltage H-bridge, primarily used to enable or disable inverter boost functions. "1" enables LVPWMx output, and "0" disables it. When the LVPWMEN enable pin is disabled, the inverter boost function will be ineffective, but the charging synchronous freewheeling function will remain effective.

8.16 Peak current protection for high-voltage side **MOSFETs**

The peak current protection circuit for the MOSFET is shown in Figure 8.14. It uses the positive terminal of the EG1160's internal comparator as the input and the negative terminal as the internal component. A 200mV reference source is used for current

detection. When setting the peak current protection value of the MOSFET, the following calculation formulas are provided for reference: Peak current protection value of MOSFET S5: Is5_peak = 200mV * (1 + R26/R25) / RS5. For example, with the parameters shown in Figure 8.14, R26 = 10K, R25 = 4.7K, and RS5 = 5mÿ, Is1_peak = 200mV * (1 + 10K/4.7K) / 5mÿ = 62.5A. Peak current protection value of MOSFET S6: Is6_peak = 200mV * (1 + R36/R30) / RS6. For example, with the parameters shown in Figure 8.14, R36 = 10K, R30 = 4.7K, and RS6 = 5mÿ, Is2_peak = 200mV * (1 + 10K/4.7K) / 5mÿ = 62.5A. MOSFET S7...

The peak current protection value is: Is7_peak = 200mV * (1 + R33/R29) / RS7. For example, with the parameters shown in Figure 8.14, R33 = 10K, R29 = 4.7K, and RS7 = 5mÿ, Is7_peak = 200mV * (1 + 10K/4.7K) / 5mÿ = 62.5A. The peak current protection value for MOSFET S8 is: Is8_peak = 200mV * (1 + R24/R23) / RS8. For example, with the parameters shown in Figure 8.14, R36 = 10K, R30 = 4.7K, and RS8 = 5mÿ, Is8_peak = 200mV * (1 + 10K/4.7K) / 5mÿ = 62.5A.

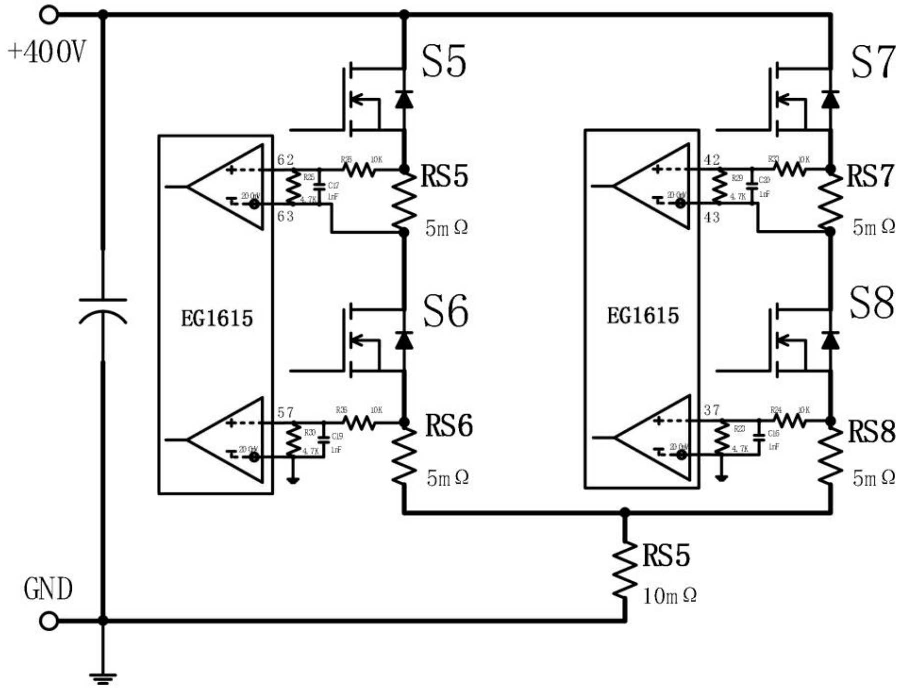

Figure 8.14 Peak current protection circuit for high-voltage side MOSFET

9. Communication function (UART)

9.1 Serial Port Description

Note: The serial port function is constantly being upgraded and updated. Please refer to the latest version of the serial communication protocol on the official website.

Serial port configuration: (9600.8.N.1) Baud

rate: 9600 Data bits: 8

Parity bits: None Stop

bits: 1

Communication

functions: The EG1615 has two serial ports: UART1 and UART2.

UART1: Primarily responsible for communication with

the EG8026. ÿ URAT1_TX: Sets the PFC voltage, current, and enable of the EG8026.

ÿ URAT1_RX: Retrieves the operating parameters of EG8026, including voltage, current, temperature, fault information, etc.

UART2: Responsible for communicating with external control panels or PC-based host computer software. It reads system information and modifies system configurations.

ÿ URAT1_TX: Sends operating parameters of EG1615 and EG8026 to the outside, including information such as voltage, current, temperature, and faults.

ÿ URAT1_RX: Receives commands from the control panel or host computer.

The serial communication function is divided into two parts: APP function and CFG function. The APP function is the normal application function, which includes the chip actively sending status messages. It also has the function of receiving external control commands. The CFG function is an advanced configuration function, mainly used to configure the chip's operating mode and calibrate parameters. The APP function is typically used when the inverter system is operating, while the CFG function is typically used when the inverter system is shut down. Parameters configured through the CFG function... The data will be stored in the FLASH space inside the chip and will be automatically loaded when the chip is powered on.

9.2 UART1

UART1 is primarily responsible for communicating with the EG8026. It sets the PFC voltage, current, and enable of the EG8026, and retrieves the EG8026's operating parameters. It includes information such as voltage, current, temperature, and faults.

9.2.1 UART1 Transmission

When the EG1615 is running, it sends a PFC control signal to the EG8026 every 1 second to control the PFC voltage, current, and enable. It is mainly used in... In charging mode, the charging current in the constant current stage and the charging voltage in the constant voltage stage are both determined by setting the PFC voltage and current.

now.

PFC configuration commands:

Byte0 Byte1 Byte2 Byte3 Byte4 Byte5 Byte6 Byte7 Byte8 Byte9 Byte10 Byte11 Byte12 Byte13 Byte14 Byte15															
0x45 0x47 0x2F 0x03 0x00					IN	VsetH VsetL IsetH			SetL	0x00	0x00	0x00	0x00	CRC16-1	

EN: PFC Enable

EN=0, PFC output is turned off, and the remaining IGBT body diodes are naturally rectified.

EN=1 enables PFC output. The PFC voltage and current are set by Vset and Iset.

Vset: PFC voltage setting

Vset is a 16-bit unsigned number, VsetH is the high byte, and VsetL is the low byte. The unit is 0.1V.

Example: Vset = 4200 means that the DC output voltage set by PFC is 420.0V.

Iset: PFC rate limiting setting

Iset is a 16-bit unsigned number, IsetH is the high byte, and IsetL is the low byte. The unit is 0.01A.

Example: Iset = 1000, indicating that the PFC current limit is set to 10.00A, and the efficiency is approximately 90%, so the effective value of the PFC current ÿ 10A/1.414*90%=6.36A

Note: The current setting for PFC cannot accurately correspond to the effective value of the AC input current, but generally speaking, the higher the set current value, the better the PCF output. The higher the upper limit of output power, the better. Because in EG8026 applications, PFC mode and inverter mode share the inductor and IGBT power transistor, therefore the PFC PWM... The frequency is 20KHz, which causes the PFC to operate in discontinuous mode at low power and in continuous mode at high power, but the current sawtooth ripple is large.

The PFC current limit value needs to be set according to the actual circuit. In our typical application, Iset=1000, and the actual effective value of the AC input current is approximately...

The current is 6.36A, and the PFC power is around 1300W.

9.2.2 UART1 Receiver

UART1 will receive APP messages from the EG8026, obtaining parameters such as voltage, current, temperature, and fault information of the EG8026.

This message is periodically sent by EG8026 and received and parsed by EG1615.

EG8026 Status Message:

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11	Byte12	Byte13	Byte14	Byte15
0x55	VacH	VacL		IacH	IacL	VdcH	VdcH	IoutH		IoutL	Tigbt	Error	Tpcb	PwrH	PwrL	CRC16

Vac: AC voltage

Vac is a 16-bit unsigned number, VacH is the high byte, and VacL is the low byte. The unit is 0.1V.

Example: Vac = 2200, indicating that the current effective value of the AC voltage is 220.0V.

Iac: Alternating Current

Iac is a 16-bit unsigned number, IacH is the high byte, and IacL is the low byte. The unit is 0.01A.

Example: Iac = 550, indicating that the current effective value of the AC current is 5.50A.

Vdc: DC bus voltage

Vdc is a 16-bit unsigned number, VdcH is the high byte, and VdcL is the low byte. The unit is 0.1V.

Example: Vdc = 3950, indicating that the current average DC bus voltage is 395.0V.

Iout: AC input/inverter output current

Iout represents the H-bridge sampling current; during inversion, it is the AC output current, and during charging, it is the AC input current. The data is represented by 2 bytes, with a minimum of...

The resolution is 0.01A.

Example: [0x02, 0xCF] uses two hexadecimal numbers to represent voltage. 0x02 is 2 in decimal, and 0xCF is 207 in decimal.

The decimal value is 2*256+207=719, and the current I is calculated as 719*0.01A=7.19A.

Tigbt: IGBT temperature

~

Tigbt is an 8-bit signed number with a range of -40 to 127. The unit is ÿ.

Example: Tigbt = 25 means the current temperature is 25ÿ.

Example: Tigbt = -40(0xD8) means the current temperature is -40ÿ.

Error: Fault code

EN (bit7): Enable off

OT (bit6): Overtemperature

RES (bit 5): Input overvoltage

RES (bit 4): Input undervoltage

OVo(bit3): Output overvoltage

UVo(bit2): Output undervoltage

OL (bit1): Overload

SC (bit0): Short circuit

Tpcb: PCB temperature

Tpcb is an 8-bit signed number, ranging from -40. For ~ 127. The unit is °C.

example, Tpcb = 25 indicates the current temperature is 25ÿ.

Example: Tpcb = -40(0xD8) means the current temperature is -40ÿ.

Pwr: Active power of alternating current

Pwr is a 16-bit unsigned number, PwrH is the high byte, and PwrL is the low byte. The unit is 1W.

Example: Pwr = 1245, indicating that the current AC output active power is 1245W.

9.3 UART2

UART2 is primarily responsible for communicating with external control panels or PC-based host computer software. It reads system information and modifies system configurations. This includes communicating with external systems.

Send operating parameters for EG1615 and EG8026, including voltage, current, temperature, and fault information. Receive commands from the panel or host computer to set...

Configure system parameters.

9.3.1 UART2 Transmission

After the chip is powered on, it will continuously send status messages to the outside at 100ms intervals, with a length of 32 bytes.

EG1615 Status Message:

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11	Byte12	Byte13	Byte14	Byte15
0x55	VacH	VacL	IacH	IacL	VdcH VdcH PrdH			PrdL	Tigbt	Error	Tpcb PwrH	PwrL VbuvH			VbuvL
Byte16	Byte17	Byte18	Byte19	Byte20	Byte21	Byte22	Byte23	Byte24	Byte25	Byte26	Byte27	Byte28	Byte29	Byte30	Byte31
VbatH	VbatH	IbatH	IbatL	VhvH	VhvL FpwmH FpwmL			State	Error	Tpcb	IhvH	IhvL	SW	CRC	

Byte1–BYTE13 are the forwarded APP messages from EG8026, see 10.2.1 for a description of USART1 reception.

Byte14–BYTE29 is the status message for EG1615.

Byte30–BYTE31 contains the CRC checksum of the first 30 bytes.

Vbat: Battery voltage

Vbat is a 16-bit unsigned number, VbatH is the high byte, and VbatL is the low byte. The unit is 0.01V.

Example: Vbat = 4880, indicating that the current average battery voltage is 48.80V.

Ibat: Battery Current

Ibat is a 16-bit signed number. The charging current is positive, and the discharging current is negative. IbatH is the high byte, and IbatL is the low byte. The unit is 0.01A.

Example: Ibat = 2200, indicating that the current battery charging current is 22.00A.

Example: Ibat = -2200, indicating that the current battery discharge current is 22.00A.

Vhv: Bus voltage feedback

Vhv is a 16-bit signed number with a value range of 0-4095, corresponding to a voltage of 0-3.3V on the VFB_HDC feedback pin.

For a description of Vhv, please refer to "8.11 VFB_HDC High-Voltage Side Bus DC Voltage Feedback".

Fpwm: PWM frequency

Fpwm is a 16-bit unsigned number, where FpwmH is the high byte and FpwmL is the low byte. The unit is 10Hz. Example: Fpwm = 7300, indicating that the current PWM frequency is 73kHz.

State: Running status

State is an 8-bit unsigned number indicating the current operating state of the EG1615. 0: Sleep mode; 1: Discharge initialization; 2: Discharge mode; 3: Charge initialization; 4: Charge mode; 5: Discharge test mode; 6: Charge test mode; 7: I/O test mode; 8: Sleep mode.

Error: Fault code (EG1615)

Error is an 8-bit unsigned number, with each bit representing a fault state.

EN (bit 7): Serial port closed

OT (bit6):HCEN

RES(bit5):LVEN

RES (bit 4): Overtemperature

OVo(bit3): Battery overvoltage

UVo(bit2): Battery low voltage

OL (bit1): Overload

SC (bit0): Short circuit

Tpcb: PCB **temperature (EG1615)**

Tpcb is an 8-bit signed integer, ranging from -40. For ~ 127. The unit is °C. example, Tpcb = 25 indicates the current temperature is 25ÿ. Similarly, Tpcb = -40 (0xD8) indicates the current temperature is -40ÿ.

Ihv: Transformer high-voltage current feedback

Ihv is a 16-bit signed number with a value range of 0-4095, corresponding to a voltage of 0-3.3V on the IFB_HV feedback pin.

For a description of IFB_HV, please refer to "8.8 IFB_HV High Voltage Overcurrent Protection".

9.4 CFG Function

The CFG (Configuration Function Group) function is an advanced configuration feature that primarily enables chip operating mode configuration and parameter calibration. The CFG function is typically used when the inverter system is shut down. Parameters configured via the CFG function are stored in the chip's internal FLASH memory and are automatically loaded when the chip is powered on.

The CFG function requires an externally sent request message. The chip responds to the request service and replies with an

acknowledgment message. Both sending and receiving use a fixed length of 16 bytes. Messages begin with ASCII codes 'E' and 'G' and end with CRC16. To distinguish between APP messages and CFG messages, the CRC checksum results are slightly different. The CRC checksum result for an APP message is f(X ₁₆ + X ₁₅ + X₂ + 1). The CRC checksum result for a CFG message is equivalent to a normal CRC checksum minus 1, i.e., the CFG message checksum result is f(X ₁₆ + X ₁₅ + X₂ + 1) - 1.

9.4.1 CFG Request Message

CFG request message format:

CFG request message (received after 50ms timeout)	
BYTE0 Header 1	0x45 - 'E'
BYTE1 Header 2	0x47 - 'G'
BYTE2 Service Code (SID)	Service content requested by the host
BYTE3 Sub-function (sfun) / Address (addr) Sub-function or address under the current service	
BYTE4 Request Data 1	
BYTE5 Request Data 2	
BYTE6 requests data 3	
BYTE7 requested data 4	
BYTE8 requested data 5	
BYTE9 requested data 6	
BYTE10 requests data 7	
BYTE11 requested data 8	
BYTE12 requests data 9	
BYTE13 requests data 10	
BYTE14 CRC checksum high byte	Cyclic redundancy check (CRC16) is calculated as **CRC16 = f(X ₁₆ + X ₁₅ + X ₂ + 1) - 1.** The first 14 bytes (BYTE0-BYTE13) are processed using CRC16, and BYTE14 is the high byte of the check result.
BYTE15 CRC checksum low byte	BYTE15 = Low byte of the verification result.

9.4.2 CFG Response Message

CFG response message format:

CFG request message (received after 50ms timeout)	
BYTE0 Header 1	0x45 - 'E'
BYTE1 Header 2	0x47 - 'G'
BYTE2 Service Code (SID) CFG Service Code requested by the host	
BYTE3 Sub-function (sfun) / Address (addr) Sub-function or address under the current service	

BYTE4 Response Data 1	
BYTE5 Response Data 2	
BYTE6 Response Data 3	
BYTE7 Response Data 4	
BYTE8 Response Data 5	
BYTE9 Response Data 6	
BYTE10 Response Data 7	
BYTE11 Response Data 8	
BYTE12 Response Data 9	
BYTE13 Response Data 10	
BYTE14 CRC checksum high byte	Cyclic Redundancy Check **(CRC16) = f(X16 + X15 + X2 + 1) - 1** Perform a CRC16 operation on the first 14 bytes BYTE0-BYTE13, where BYTE14 equals the high byte of the check result.
BYTE15 CRC checksum low byte	BYTE15 = Low byte of the verification result.

9.4.3 0x10 Service - Session Switching

The 0x10 service is a session switching service. Communication sessions are mainly divided into default session (01), extended session (03), and programming session (02).

The programming session is not currently available to users.

Host 0x10 Request Message:

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11	Byte12	Byte13	Byte14	Byte15
0x45	0x47	0x10	Session	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	CRC16-1	

addr is the address of the DID; different addresses store different DID information.

Slave reply:

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11	Byte12	Byte13	Byte14	Byte15
0x45	0x47	0x10	Session	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	CRC16-1	

If the slave device replies with Byte4=0xCC, it indicates that the session switch was successful; if it replies with Byte4=0xEE, it indicates that the session switch failed.

Default session (Session = 01):

In the default session, the slave device periodically sends APP messages. Communication is performed in the default session by default.

Programming Session (Session = 02):

This is used when the program returns to the bootloader after flashing; it is not currently available to users.

Extended Session (Session = 03):

In extended sessions, the slave device stops periodically sending APP messages. This is typically done when DID or CFG read/write operations are required, to avoid APP messages consuming bandwidth.

Use communication resources to switch to the extended session; after the operation is complete, switch back to the default session.

Pass-through (Session = 04):

When Byte4 = '8', Byte5 = '0', Byte6 = '2', and Byte7 = '6', serial port data will be transparently transmitted to the EG8026, allowing further processing of the EG8026.

Line read and write operations.

When Byte4 = '1', Byte5 = '6', Byte6 = '1', and Byte7 = '5', the serial port data will be transparently transmitted back to the EG1615, and the serial port can be re-connected.

EG1615 is used for read and write operations.

9.4.4 0x22 Service - Read DID

The 0x22 service is a DID read service. System configuration parameters, version information, etc., are stored in the DID. By requesting the 0x22 service, the host can...

This allows you to read the chip's configuration parameters and version information.

Host 0x22 Request Message:

Byte0 Byte1	Byte2 Byte3	Byte4 Byte5	Byte6 Byte7	Byte8 Byte9	Byte10 Byte11	Byte12 Byte13	Byte14 Byte15								
0x45 0x47	0x22 addr	0x00 0x00 0x00	0x00 0x00 0x00							0x00	0x00	0x00	0x00	CRC16-1	

addr is the address of the DID; different addresses store different DID information.

Slave reply:

Byte0 Byte1	Byte2 Byte3	Byte4 Byte5	Byte6 Byte7	Byte8 Byte9	Byte10 Byte11	Byte12 Byte13	Byte14 Byte15								
0x45 0x47	0x22 addr			d1	d2	d3	d4	d5	d6	d7	d8	d9	d10	CRC16-1	

If the slave device replies with d1~d10 all being 0xFF, it means that reading the DID failed.

9.4.5 0x2E Service - Write DID

The 0x2E service is the DID write service. By requesting the 0x2E service, the host can write configuration parameters and version information into the chip.

Host 0x2E Request Message:

Byte0 Byte1	Byte2 Byte3	Byte4 Byte5	Byte6 Byte7	Byte8 Byte9	Byte10 Byte11	Byte12 Byte13	Byte14 Byte15								
0x45 0x47	0x2E addr			d1	d2	d3	d4	d5	d6	d7	d8	d9	d10	CRC16-1	

addr is the address of the DID; different addresses store different DID information.

Slave reply:

Byte0 Byte1	Byte2 Byte3	Byte4 Byte5	Byte6 Byte7	Byte8 Byte9	Byte10 Byte11	Byte12 Byte13	Byte14 Byte15								
0x45 0x47	0x2E addr			resp	0x00 0x00	0x00 0x00 0x00				0x00	0x00	0x00	0x00	CRC16-1	

resp = 0xCC : Write successful

resp = 0xEE: Write failed

DID Information Table:

DID	ADDR D4 D5 D6 D7 D8					D9 D10 D11 D12 D13 CRCH CRCL							
Inverter configuration: 0x04 Reserve		cfg	·	·	·	·	·	·	·	·	·	CRC-1	WR
battery	0x05 Reserved					·	·	·	·	·	·	CRC-1	WR
voltage calibration value; 0x06 Reserve		VbatAdj				·	·	·	·	·	·	CRC-1	WR
battery current calibration value; 0x07		IlvAdj				·	·	·	·	·	·	CRC-1	WR
Reserve	0x08 Reserved					·	·	·	·	·	·	CRC-1	WR
production date	0x09	YY	YY MM DD			·	·	·	·	·	·	CRC-1	WR
serial number.	0x0A EG20010001											CRC-1	WR

Part Number,	0x0B DC48V2000W	CRC-1	WR
Customer Code,	0x0C C017	CRC-1	WR
Kernel Version,	0x0D EG1615 v10	CRC-1	WR
Software Version Number, Hardware Version	1.0.220228	CRC-1	R
Number , Checksum	1.0.220228	CRC-1	R
	0x11 0x1234	CRC-1	R

9.4.6 0x21 Service - Read CFG

Service 0x21 is the CFG read service, which handles inverter operation and protection parameters. By requesting service 0x21, the host can read the chip's inverter operation parameters.

Line, protection parameters, and other content.

Host 0x21 Request Message:

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11	Byte12	Byte13	Byte14	Byte15
0x45	0x47	0x21	addr	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	CRC16-1	

addr is the address of the CFG; different addresses store different CFG information.

Slave reply:

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11	Byte12	Byte13	Byte14	Byte15
0x45	0x47	0x21	addr	d1	d2	d3	d4	d5	d6	d7	d8	d9	d10	CRC16-1	

If the slave device replies with d1~d10 all being 0xFF, it indicates that reading CFG failed.

9.4.7 0x2D Service - Write CFG

The 0x2D service is a write CFG service. By requesting the 0x2D service, the host can write configuration parameters and version information to the chip.

Host 0x2D Request Message:

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11	Byte12	Byte13	Byte14	Byte15
0x45	0x47	0x2D	addr	d1	d2	d3	d4	d5	d6	d7	d8	d9	d10	CRC16-1	

addr is the address of the CFG; different addresses store different CFG information.

Slave reply:

Byte0	Byte1	Byte2	Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10	Byte11	Byte12	Byte13	Byte14	Byte15
0x45	0x47	0x2D	addr	resp	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	0x00	CRC16-1	

resp = 0xCC : Write successful

resp = 0xEE: Write failed

CFG Information Sheet:

CFG	ADDR	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	CRC H	CRC L	
Save parameters	0x00	·	·	·	·	·	·	·	·	·	·	CRC-1		WR
Factory reset 0x01		0x88	0x20	0x55	0x36	·	·	·	·	·	·	CRC-1		WR
Register configuration generated 0x04		·	Active	·	·	·	·	·	·	·	·	CRC-1		WR

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			and										
Effective	0x05											**CRC-1**	**WR**
retention	0x06											**CRC-1**	**WR**
retention	0x07											**CRC-1**	**WR**
retention	0x08											**CRC-1**	**WR**
retention charging settings	0x10	Fpwm		Chg_Vset		Chg_Timer		VSet_Pfc		ISet_Pfc		**CRC-1**	**WR**
Synchronous rectification setting 0x11		RectOn_Iref		RectOff_Iref		Rect_DutyStart		Rect_DutySte p				**CRC-1**	**WR**
Discharge setting	0x12	Freq		LowFreq		Chg_d t	Dis_d t	Freq_Iref		LowFreq_Iref		**CRC-1**	**WR**
Hiccup setting	0x13	HicOn_Vref		HicOff_Vref		HicOff_Iref						**CRC-1**	**WR**
Short circuit protection	0x20	Ref		Delay		Recover_Ref		Recover_Dela and		Recover_Time s		**CRC-1**	**WR**
Overload protection	0x21	Ref		Delay		Recover_Ref		Recover_Dela and		Recover_Time s		**CRC-1**	**WR**
Overvoltage protection	0x22	Ref		Delay		Recover_Ref		Recover_Dela and		Recover_Time s		**CRC-1**	**WR**
Undervoltage protection	0x23	Ref		Delay		Recover_Ref		Recover_Dela and		Recover_Time s		**CRC-1**	**WR**
Over-temperature protection	0x24	Ref		Delay		Recover_Ref		Recover_Dela and		Recover_Time s		**CRC-1**	**WR**
Input overcurrent protection 0x25		ChgRef		Recover_ChgRef		DisChgRef		Recover_DisChgR if				**CRC-1**	**WR**
Inverter control parameters 0x26			cfg									**CRC-1**	**WR**

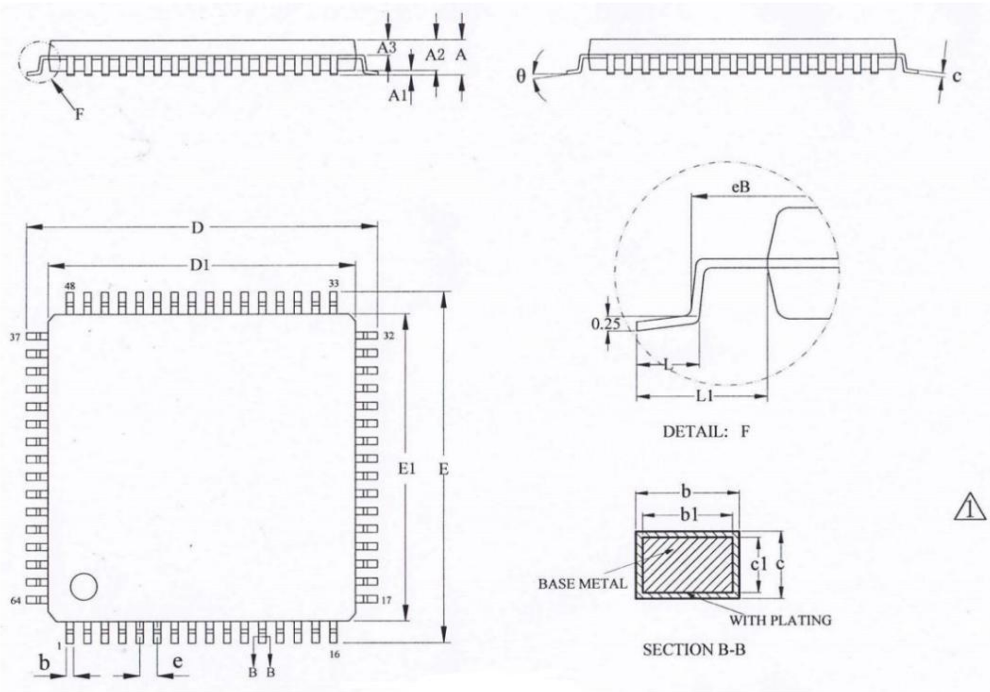
CFG Parameter Description Table:

address	**Parameter name**	**symbol**	**data type**	**unit**	**Conversion coefficient**	**Example** Serial port value	actual value	**Pin settings**
0x00	**Save all**							
0x01	**Factory reset**							
0x04 Register configuration is valid		RegCfgEnable	int16			1. Internal		
0x10	Charging mode PWM frequency Chg_Freq charging		int32	Hz	1	73000	73KHz **FADJ**	
	voltage setting	Chg_Vset	int16	In	0.1	588	58.8V **PISTON**	
	Charging mode timer	Chg_Timer	int16	min	1	180	3hour	
	PFC voltage setting	Chg_Pfc_Vset	int16	In	0.1	4200	420V	
	PFC current setting value	Chg_Pfc_Iset	int16	A	0.01	1000	10A **CHG_ISET**	
0x12	Discharge mode PWM frequency DisChg_Freq Discharge		int32	Hz	1	73000	73KHz **FADJ**	
	mode PWM frequency 2	DisChg_LowFreq	int32	Hz	1	65000	65KHz **FADJ-10KHz**	

	Charging mode dead zone	Chg_Deadtime	you8	us	1/60	30	0.5us	**DT1/DT0**
	discharge mode dead zone	DisChg_Deadtime	you8	us	1/60	30	0.5us	**DT1/DT0**
	discharge frequency switching	DisChg_Freq_Iref	int16	A	0.01	800	8A	
	point discharge frequency recovery point	DisChg_LowFreq_Iref	int16	A	0.01	600	6A	
0x21	Overload protection reference	OL_Ref	int16	mV	1	650	650mV	
	Overload protection delay	OL_Delay	int16	mS	1	10000	10S	
	Overload protection recovery benchmark	OL_Recover_Ref	int16	mV	1	550	550mV	
	Overload protection recovery delay	OL_Recover_Delay	int16	mS	1	10000	10S	
	Overload protection recovery times	OL_Recover_Times	int16	times	1	5	5 times	
0x22	Overvoltage protection	OV_Ref	int16	In	0.1	650	65.0V	**PISTON**
	reference, overvoltage	OV_Delay	int16	mS	1	10000	10S	
	protection delay, overvoltage protection	OV_Recover_Ref	int16	In	0.1	600	60.0V	
	recovery reference, overvoltage protection	OV_Recover_Delay	int16	mS	1	10000	10S	
	recovery delay, overvoltage protection recovery times.	OV_Recover_Times	int16	times	1	5	5 times	
0x23	Undervoltage protection reference	UV_Ref	int16	In	0.1	420	42.0V	
	Undervoltage protection delay	UV_Delay	int16	mS	1	10000	10S	
	Undervoltage protection recovery benchmark	UV_Recover_Ref	int16	In	0.1	450	45.0V	
	Undervoltage protection recovery delay	UV_Recover_Delay	int16	mS	1	10000	10S	
	Undervoltage protection recovery times	UV_Recover_Times	int16	times	1	5	5 times	
0x24	Over-temperature protection	OT_Ref	int16	ÿ	1	85	85ÿ	
	reference, over-temperature	OT_Delay	int16	mS	1	2000	2S	
	protection delay, over-temperature protection	OT_Recover_Ref	int16	ÿ	1	75	75ÿ	
	recovery reference, over-temperature protection recovery delay	OT_Recover_Delay	int16	mS	1	10000	10S	
	Over-temperature protection recovery times	OT_Recover_Times	int16	times	1	5	5 times	
0x25	Charging protection point	ChgRef	int16	A	0.01	5000	50A	
	Charging recovery point	Recover_ChgRef	int16	A	0.01	4500	45A	
	Discharge protection point	DisChgRef	int16	A	0.01	6000	60A	
	Discharge recovery point	Recover_DisChgRef	int16	A	0.01	5500	55A	
0x26	Inverter Configuration Page Selection	Cfg	int16	-	-	Configuration 1		
0x27	Switch Mode Configuration	Type	int16	-	-	0/1	Button/ switch	

10. Package size

10.1 LQFP64



SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	—	—	1.60
A1	0.05	—	0.15
A2	1.35	1.40	1.45
A3	0.59	0.64	0.69
b	0.18	—	0.26
b1	0.17	0.20	0.23
c	0.13	—	0.17
c1	0.12	0.13	0.14
D	11.80	12.00	12.20
D1	9.90	10.00	10.10
E	11.80	12.00	12.20
eB	11.05	—	11.25
E1	9.90	10.00	10.10
e	0.50BSC		
L	0.45	—	0.75
L1	1.00REF		
θ	0	—	7°